

(1) Right Angled Triangle Pattern Using Stars.

Ans) rows = int(input("Enter the Number of Rows:"))

for i in range(1, rows + 1):

print("*" * i)

(2) Reverse a String Without Using Built-in Functions:-

Ans) input_str = input("Enter a String: ")

reversed_str = ""

for char in input_str:

reversed_str = char + reversed_str

print("Reversed String:", reversed_str)

(3) Check if a Number is a palindrome:-

num = int(input("Enter a number:"))

Original = num

reverse = 0


```
while num > 0:
```

```
    digit = num % 10
```

```
    reverse = reverse * 10 + digit
```

```
    num = num // 10
```

```
if original == reverse:
```

```
    print(f"{original} is a palindrome.")
```

```
else
```

```
    print(f"{original} is a non-palindrome.")
```

4) Print Number in Ascending and Descending Order Using a Loop:-

Ans) `n = int(input("Enter the Value of n:"))`

```
Print ("Ascending Order:")
```

```
for i in range(1, n+1):
```

```
    print(i, end=" ")
```

```
Print ("\n Descending Order:")
```

```
for i in range(n, 0, -1):
```

```
    Print(i, end=" ")
```


(5) Sum of Squares of First n Natural Numbers

Ans) $n = \text{int}(\text{input}(\text{"Enter the Number of Values : "}))$

$\text{Sum_Squares} = 0$

for i in $\text{range}(1, n+1)$:

$\text{Sum_Squares} += i * i$

$\text{Print}(f\text{"Sum of Squares of first \{n\} natural Numbers is \{Sum_Squares\}. "})$.